

Claude Code for Academic Researchers

A Methods Workshop in AI-Assisted Research

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Research: Political economy of technological change

- Automation, AI, and political behavior
- LLM-based text classification pipelines

Background:

- M.Res. Political Science (UPF)
- M.Sc. Economics (Barcelona School of Economics)
- Former data scientist

Why this course:

- Daily user of Claude Code for academic research
- Use it across the full pipeline: data cleaning, analysis, robustness checks, code auditing, replication, writing, reviewing
- These tools can dramatically expand what a researcher can accomplish, if used with the right workflows

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- ➊ Who are you? (name, department, research area)
- ➋ What does your daily workflow look like? (tools, languages, data)
- ➌ How do you currently use AI, if at all?
- ➍ What do you hope to learn from this course?

Course Roadmap

Session	Modules	Date
1	1. What This Is & What You Can Do With It 2. How Claude Code Works	Today
2	3. Automating Research Tasks 4. Robustness and Replication	Thu Apr 9, 11am–1pm
3	5. Code Auditing and Review 6. Version Control with Git	Fri Apr 17, 11am–1pm
4	7. Developing Expert Judgment 8. The Anthropic API 9. Capstone Project	Fri Apr 24, 11am–1pm

4 sessions × 2 hours | 60–70% hands-on | Pair programming throughout

Possible 5th session: Thu Apr 30, 11am–1pm, depending on participants' needs

The Central Idea

You are the principal investigator. Claude Code is the research assistant. You direct the work, make analytical decisions, and verify every result. Claude Code handles the mechanical execution.

The question is never “can Claude Code do this.” It almost always can.

The question is: **what is the most productive and reliable way to get this done?**

By the end of this course, you will be able to:

- Automate data cleaning, robustness checks, code auditing, and documentation
- Choose the right execution mode for each task
- Verify output against known properties of your data
- Recognize when Claude Code is failing and intervene effectively

Module 1

What This Is and What You Can Do With It

Demo + Exercise

Get the Course Code

Open your terminal and run:

```
git clone https://github.com/bradyallardice/claude-code-workshop.git
```

This downloads the course repository to your computer. Then open the folder in VS Code:

File → **Open Folder** → select claude-code-workshop

Already cloned it? Just run `git pull` from inside the folder to get the latest version.

Setup Check

Open the terminal in VS Code (or your standalone terminal) and run:

- 1 Verify Node.js: `node --version`
- 2 Verify Claude Code: `claude --version`
- 3 Launch Claude Code: `claude`

You should see Claude Code's interface appear in your terminal.

Try typing a simple message: What files are in this directory?

If you completed pre-work: help your neighbor.

If you're stuck: pair with someone who is set up. We'll troubleshoot at break.

Dataset 1: County Presidential Election Returns

- Source: MIT Election Data + Science Lab
- County-level vote totals by candidate and party, 2000–2024
- ~94,000 rows, one row per county $\times$ year $\times$ candidate
- Variables: FIPS codes, candidate names, party, vote counts, vote mode

Dataset 2: IPUMS American Community Survey Microdata

- Source: IPUMS USA (University of Minnesota)
- Person-level survey records with demographic, economic, and geographic variables
- ~15 million rows per year, one row per person
- Variables: age, education, employment, income, race, poverty status, county FIPS

Goal: Build a county-year panel joining election results with demographics, and visualize the outputs.

This requires:

- ➊ Reshape election data to one row per county-year with vote shares
- ➋ Aggregate 15M person-level records to county-level means using survey weights
- ➌ Harmonize FIPS codes across datasets (different formats)
- ➍ Merge on county $\times$ year and validate the join
- ➎ Create visualizations of trends over time

In Stata or R, this takes 30–45 minutes of careful coding.

What Just Happened

It wrote the code:

- Read both files
- Identified the join key
- Handled cleaning steps
- Wrote a complete script

It also:

- Understood the data structure (person-level vs. county-level, survey weights, FIPS formats)
- Ran the code and diagnosed errors
- Provided summaries of the output (row counts, variable descriptions, merge diagnostics)

But it also made a decision about our data without asking.

Both of these things are true:

- 1 It took 90 seconds instead of 45 minutes
- 2 It made analytical choices autonomously

How to Think About Claude Code

Claude Code is like a research assistant who is:

- Extremely fast
- Has read a lot
- Will **never** say “I don’t know how to do that”

Instead, it will just do it wrong and hand it in.

You have all managed RAs. You know how valuable, and how dangerous, that profile is.

This course teaches you to capture the value while managing the risk.

The Coding Paradigm Spectrum: Manual Coding

Manual Coding

Autocomplete / Copilot

Web Chat

Agentic Coding

You write every line. Full control, no assistance.

This is what you do now in Stata or R.

```
def load_tasks_to_dwas(tasks_to_dwas_file: str) -> pd.DataFrame:
    """
    Load Tasks-to-DWAs mapping file.

    Returns DataFrame with columns: O*NET-SOC Code, Task ID, DWA ID, DWA Title
    """
    if not os.path.exists(tasks_to_dwas_file):
        raise FileNotFoundError(f"Tasks-to-DWAs file not found: {tasks_to_dwas_file}")

    df = pd.read_excel(tasks_to_dwas_file)

    required_cols = ['O*NET-SOC Code', 'Task ID', 'DWA ID', 'DWA Title']
    missing = [c for c in required_cols if c not in df.columns]
    if missing:
        raise ValueError(f"Missing columns in Tasks-to-DWAs file: {missing}. Available: {list(df.columns)}")

    logger.info(f"Loaded {len(df)} task-DWA mappings ({df['Task ID'].nunique()} unique tasks, "
                f"{df['DWA ID'].nunique()} unique DWAs)")

    return df

def load_dwa_reference(dwa_reference_file: str) -> pd.DataFrame:
    """
    Load DWA Reference file with IWA (Intermediate Work Activity) context.

    Returns DataFrame with columns: DWA ID, DWA Title, IWA Title, Element Name
```

The Coding Paradigm Spectrum: Autocomplete / Copilot

Manual Coding

Autocomplete / Copilot

Web Chat

Agentic Coding

```
1 // create a class in TypeScript to represent a student that has a name, an id, and a list of courses
2 // the class should have method to add and remove a course to the list of courses
3
4 class Student {
  name: string;
  id: number;
  courses: string[];

  constructor(name: string, id: number, courses: string[]) {
    this.name = name;
    this.id = id;
    this.courses = courses;
  }

  addCourse(course: string) {
    this.courses.push(course);
  }

  removeCourse(course: string) {
    this.courses = this.courses.filter((c) => c !== course);
  }
}
```

AI suggests the next line as you type.

You accept or reject. The AI assists; you drive.

The Coding Paradigm Spectrum: Web Chat

Manual Coding

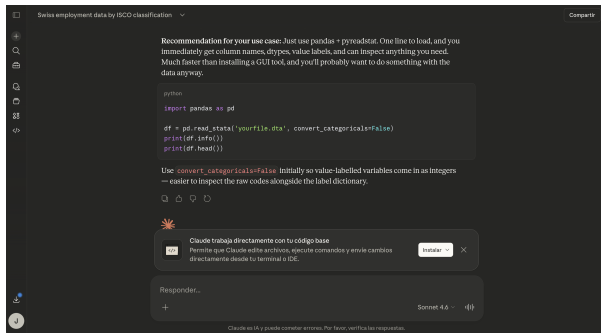
Autocomplete / Copilot

Web Chat

Agentic Coding

You describe a task; AI gives you code to copy-paste.

The AI cannot see your project.



The Coding Paradigm Spectrum: Agentic Coding

Manual Coding

Autocomplete / Copilot

Web Chat

Agentic Coding

AI reads, writes, and runs code **in your project**.

- So 69 fuzzy + 241 low-score token = **310 borderline matches** the LLM confirmed but we're not fully confident in. Let me flag those as `needs_review=True`:

- **Read** `improve_shp_firm_matching.py` (lines 1639-1663)

- **Edit** `improve_shp_firm_matching.py`

Added 10 lines

```
□ # --- Flag borderline matches as needs_review even if LLM confirmed ---
□ # fuzzy_char and low-score token matches are inherently less reliable
□ borderline_mask = (
□     (df["match_method"] == "fuzzy_char") |
□     ((df["match_method"].isin(["token_exact", "token_relaxed"]) & (df["match_score"]
□     <= 0.50))
□ )
□ n_borderline_flagged = ((borderline_mask) & (~df["needs_review"])).sum()
□ df.loc[borderline_mask, "needs_review"] = True
□ logger.info(f" Borderline flagged: {n_borderline_flagged} matches set to
□ needs_review=True")
```

- **Bash** Re-compile with borderline flagging

```
IN  python3 improve_shp_firm_matching.py compile --output-dir Data/shp_matching/ 2>&1

OUT 2026-03-05 21:53:32 [INFO] Loaded 11780 raw matches from matched_names_raw.csv
    2026-03-05 21:53:32 [INFO] LLM rejected: removed 532 matches
    2026-03-05 21:53:32 [INFO] Manual verdicts: removed 4, confirmed 10
```

- **Final: 11,244 matched** — 10,842 confirmed, 402 flagged `needs_review=True`.

The 402 flagged entries are:

Why Agentic Coding Is Different

Paradigm	Pros	Cons
Manual Coding	Full control	Slow, everything by hand
Autocomplete / Copilot	You control logic and structure	Still line-by-line
Web Chat	Can handle complex tasks	Copy-paste; AI misses context
Agentic Coding	Fastest; works on your project	Can act autonomously

The Key Insight

Agentic coding is the first paradigm where the tool **interacts with your project directly and autonomously**. That is what makes it so productive, and why learning to use it well matters.

These principles apply to Claude Code, Cursor, Codex, Windsurf, or whatever comes next.

What Claude Code Can Do for Academic Research

Data Work

- Clean, reshape, and merge datasets
- Construct variables from raw data
- Build panel datasets across sources
- Handle missing data, recoding, harmonization

Analysis

- Exploratory data analysis and summary statistics
- Generate robustness checks and specification variants
- Create publication-quality visualizations
- Synthesize results across specifications

Writing and Review

- Draft methods sections, data descriptions, results summaries
- Audit existing scripts for silent errors
- Review code against your methods section
- Write documentation, READMEs, replication packages

Research Infrastructure

- Process text corpora at scale (NLP, classification)
- Web scraping and API data collection
- Crosswalk and linkage construction
- Refactor and optimize slow pipelines

A Tool Built for Software Engineers

Claude Code was designed for professional developers. Their world is different from ours.

Software Engineering

- “Works” = **runs correctly**. Clear success criteria: tests pass, feature ships, users are happy
- Code is the product
- Errors are caught by automated tests
- Speed and iteration are paramount

Academic Research

- “Works” = **???**. Code that runs without errors can produce completely wrong results. Output that looks plausible can be based on the wrong sample, wrong specification, or wrong variable
- Code is the *instrument*; the paper is the product
- Errors are caught by *you*, or by Reviewer 2
- Correctness matters more than speed

The defaults, instincts, and shortcuts built into these tools reflect engineering priorities.

Using them well for research means overriding some of those defaults.

Do You Own It?

Pick any result your code produced. Without looking at the code: what got dropped, what got merged, and why?

If you can't answer that in 30 seconds, you don't own it.

Before you delegate:

- What would wrong look like? Can I describe a specific failure mode before I run this?

After you delegate:

- Can I describe what this code does without reading it?
- Can I state what decisions shaped the output and why?

You didn't write it. That's fine.

But every decision in it is yours. **Can you justify it as if you chose it yourself?**

Why Verification Matters

What happens when you skip verification:

- ❶ **Silent observation loss:** A merge silently drops 300 observations.
The code looks clean. The error is invisible without checking row counts.
- ❷ **Invalid transformation:** A log transformation applied to a variable containing zeros.
No error raised, wrong results.
- ❸ **Incorrect panel operations:** A lagged variable computed without sorting by time within panels.

These are the kinds of errors that end up in published work.

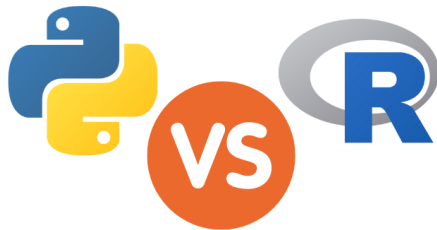
You Are the PI. Claude Code Is the RA.

The question is not “can Claude Code do this” but
“what is the best way to get this done reliably.”

If you cannot verify a result, you should not delegate the task.

Why Python

- Claude Code is substantially better at Python than Stata or R
- The principles (task decomposition, verification, control) transfer to any language
- Python is increasingly where the tools and ecosystem are for computational research

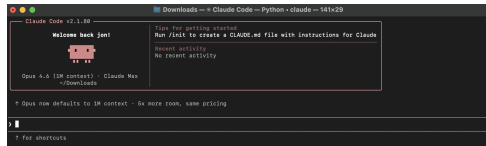


Where Does Claude Code Live?

Claude Code runs **inside your terminal**, not in a browser, not in an app.

- 1 You open a terminal
- 2 Navigate to your project: `cd my_project/`
- 3 Type `claude` and press Enter
- 4 You're now talking to Claude Code

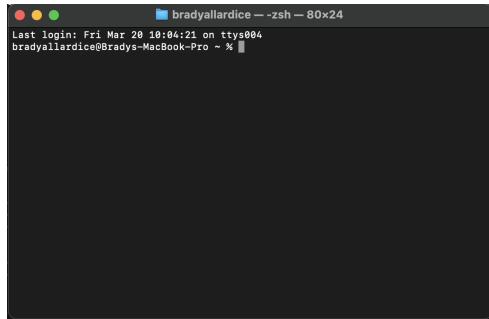
It can see your files, read your data, write scripts, and run them, all without copying and pasting.



This is what Claude Code looks like when you launch it.

What Is the Terminal?

The terminal is a text-based interface to your computer. Instead of clicking, you type commands.

A screenshot of a terminal window. The title bar shows three colored circles (red, yellow, green) and the text "bradyallardice — zsh — 80x24". The terminal content shows "Last login: Fri Mar 20 10:04:21 on ttys004" followed by the prompt "bradyallardice@Bradys-MacBook-Pro ~ %".

```
bradyallardice — zsh — 80x24
Last login: Fri Mar 20 10:04:21 on ttys004
bradyallardice@Bradys-MacBook-Pro ~ %
```

Claude Code writing and running code in the terminal.

Essential commands:

Where am I?

```
pwd
```

What's in this folder?

```
ls
```

Move into a folder

```
cd my_project/
```

Run a Python script

```
python my_script.py
```

Launch Claude Code

```
claude
```

Why VS Code?

VS Code is a free code editor with a built-in terminal.

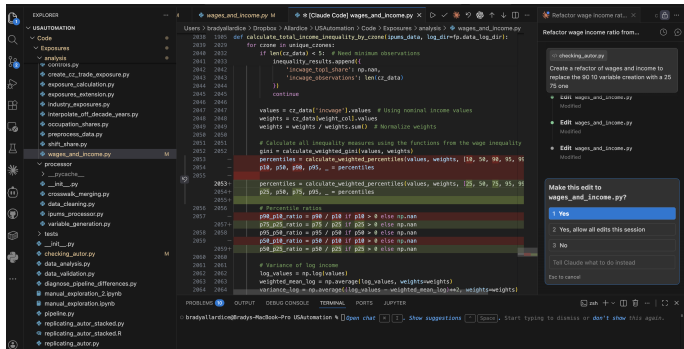
Everything in one window:

- **Left:** file explorer
- **Center:** your code
- **Right:** Claude Code

You see what Claude Code writes *as it writes it*.

Today we use three features:

- 1 Open a folder
- 2 View files in the editor
- 3 Talk to Claude Code



Coming from RStudio? ▶ Configure VS Code to look like RStudio

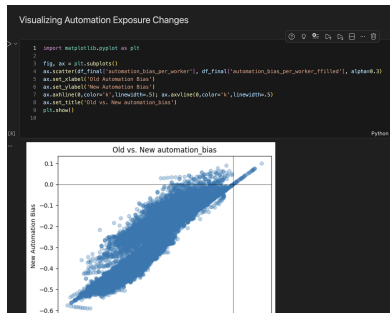
Python Files: .py vs. .ipynb

Notebooks (.ipynb)

```
1 df_ipms_pre_exposure = pd.read_pickle('~/Users/bradyallardice/Desktop/PHD/Projects/95Automation/Data/Merge/EPWG/processed_ipms_1948.pkl')
2 df_ipms_pre_exposure.groupby(['occ1950rj', 'case'])['perwt_by_factor'].sum().reset_index()
```

occ1950rj	case	perwt_by_factor
0	2.0	3760.0
1	2.0	3860.0
2	2.0	4200.0
3	2.0	4301.0
4	2.0	4702.0
...	...	...
68909	970.0	39206.0
68910	970.0	39301.0
68911	970.0	38902.0
68912	970.0	38903.0
68913	970.0	38400.0

68914 rows x 3 columns



Scripts (.py)

```
Code > Exposures > checking_autor.py > ...
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5 from scipy.stats import pearsonr, spearmanr
6 import os
7 import pickle
8 from pathlib import Path
9
10 # Set up plotting style
11 plt.style.use('ggplot')
12 sns.set_context('talk')
13
14 import General.filepaths as fp
15
16 def load_data_files():
17     """Load all intermediate data files needed for diagnostics"""
18     data_files = {}
19
20     # Try to load cz-level data
21     try:
22         occ_exposure_path = f'{fp.exposures_out_dir}occ_ind_4880rj_panel_cw_t85.dta'
23         data_files['occ_exposure'] = pd.read_stata(occ_exposure_path)
24         data_files['occ_exposure'] = data_files['occ_exposure'][['year', 'occ1950rj', 'N']]
25         print(f"Loaded occupation exposure data from {occ_exposure_path}")
26         print(f"Shape: {data_files['occ_exposure'].shape}")
27     except FileNotFoundError:
28         print(f"Could not find occupation exposure data at {occ_exposure_path}")
29
30     # Try to load occupation shares data
31     try:
32         occ_shares_path = f'{fp.exposures_out_dir}final_occ_shares_all_years_by_cz.csv'
33         data_files['occ_shares'] = pd.read_csv(occ_shares_path)
34         print(f"Loaded occupation shares data from {occ_shares_path}")
35         print(f"Shape: {data_files['occ_shares'].shape}")
36     except FileNotFoundError:
37         print(f"Could not find occupation shares data at {occ_shares_path}")
38
```

Why Scripts for This Course

Notebooks (.ipynb)

- Cells you run one at a time
- Great for exploration and teaching
- State depends on which cells you ran and in what order
- Hard for Claude Code to work with because it cannot “click” cells

Scripts (.py)

- Plain text file, runs top to bottom
- Reproducible: `python my_script.py` always does the same thing
- Claude Code can read, write, and run them directly
- **This is what we use today**

The Key Difference

Claude Code operates on files, not interactive sessions. Scripts run end-to-end and produce the same result every time, exactly what you need for reproducible research.

What Changes When You Use Scripts

You can run scripts line by line in VS Code (Shift+Enter), just like a notebook. But scripts aren't really built for that.

Their strength is that they run **end-to-end**: `python my_script.py` always does the same thing, top to bottom. That is what makes them reproducible, and what Claude Code needs.

What this means for your workflow:

- More **checkpointing**: save intermediate datasets to disk so you can inspect them
- More **logging**: print row counts, shapes, and summaries as the script runs
- More **intermediate file saves**: write CSVs between pipeline stages
- Overall **less interactive**, more structured

This feels slower at first. But it produces pipelines that are reproducible, debuggable, and that Claude Code can work with directly.

What is a virtual environment?

A self-contained folder that holds a separate copy of Python and its packages. Each project gets its own environment, so packages from one project don't interfere with another.

Why you need one:

- Without it, every project shares the same packages, and updating one project can break another
- Claude Code installs packages when it needs them. You want that isolated, not touching your system Python
- Reproducibility: you know exactly which packages your project uses

Think of it like this: a virtual environment is a clean desk for each project. You only put the tools you need on it, and cleaning up one desk does not affect the others.

Setting Up a Virtual Environment

Run these commands in your terminal, from your project folder:

```
# 1. Create the environment (once per project)
```

```
python3 -m venv AIAgents
```

```
# 2. Activate it (every time you open a new terminal)
```

```
source AIAgents/bin/activate # Mac/Linux
```

```
AIAgents\Scripts\activate # Windows
```

```
# 3. Install packages
```

```
pip install pandas numpy statsmodels
```

```
# 4. Confirm it's active (you should see (AIAgents) in your prompt)
```

```
which python
```

In VS Code: once you create the AIAgents folder, VS Code will detect it and ask to use it as your interpreter. Say yes. After that, every terminal it opens will activate the environment automatically.

Exercise: Your First Commands

Task 1:

Here is a Python script that cleans a dataset.

Ask Claude Code to explain what it does, line by line.

Task 2:

Ask Claude Code to add verification checks, documentation, and print statements to the script.

Task 3:

Ask Claude Code to change the two-party vote shares to total vote shares.

- What did it get right?
- Anything wrong or confusing?
- That is normal and expected

By the end of this course, you will have:

- Built a documented data pipeline
- Generated robustness checks
- Audited code for errors
- Learned version control for agent-assisted work

You will know which parts Claude Code should handle and which require your judgment.

And you will be able to take these workflows back to your own research.

Now that you have seen what Claude Code can do,
let us look at how it actually works.

Context, modes, and the conventions that make it reliable.

What you do now (Stata / R):

- Open a session, load data into memory
- Run commands interactively, see results immediately

How Claude Code works:

- Reads and writes **files on disk**
- No persistent session with your data loaded
- Reads your files → writes/modifies scripts → optionally runs them

Think in Projects, Not Sessions

You need to think in terms of:

- Project directories
- File paths
- Scripts that run end-to-end

Not interactive exploration.

The Context Window

Claude Code can only “see” a limited amount of text at once.

The context window includes:

- Your instructions
- The files it reads
- Its own responses

When a conversation gets long or a project gets large, it loses earlier context.

Why the Context Window Matters

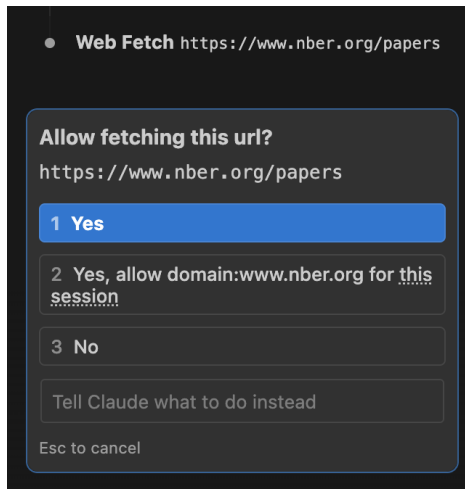
The Consistency Problem

In a complex project with many files, Claude Code may change one file in a way that is **inconsistent** with another file it can no longer see.

Errors compound when the agent loses track of earlier decisions.

We will cover strategies for managing this in Module 3.

When Claude Code Asks for Permission



Before taking actions (running code, writing files), Claude Code asks you.

Your three options:

- 1 **Yes:** allow this one time
- 2 **Yes, always allow:** allow this type of action for the rest of the session (no more prompts)
- 3 **No:** reject, and tell Claude Code what to do differently

"No" is not a dead end.

Use it to redirect: *"No, use a left join instead"* or *"No, don't overwrite that file, create a new one."*

Common Actions and Risk Levels

Action	Example you'll see	What it does	Risk level
Read	Read scripts/clean.py	Looks at a file	Safe
Glob	Glob data/**/*.csv	Finds files by name	Safe
Grep	Grep "county_fips"	Searches inside files	Safe
Edit	Edit clean.py line 42	Changes existing code	Review first
Write	Write scripts/merge.py	Creates a new file	Review first
Bash	python scripts/clean.py	Runs a command	Read before approving
Bash	pip install pandas	Installs a package	Read before approving
Bash	rm data/old_panel.csv	Deletes a file	Read before approving
Bash	mv output.csv data/	Moves a file	Read before approving
Bash	git push	Uploads to remote server	Read before approving

Safe to “always allow”:

Reading and searching. Claude Code is just looking, not changing anything.

Always read before approving:

Bash commands can do *anything*: run code, delete files, install software. Check what it is actually running.

Protecting Sensitive Files

“Always allow reads” assumes there is nothing sensitive to read.

If your project has restricted data or IRB-protected files, you need to hide them.

How: Create a plain text file called `.claudeignore` in your project folder.

Each line is a pattern. Any file matching a pattern becomes **invisible** to Claude Code. It cannot read, search, or reference it.

Example `.claudeignore`:

```
# Everything in the restricted folder
data/restricted/

# Any file whose name starts with "identifiable_"
data/raw/identifiable_*

# A specific file
data/raw/ssn_crosswalk.csv
```

Setting Up .claudeignore

- 1 Open your project folder in VS Code
- 2 Create a new file named exactly `.claudeignore`
(note the dot at the start, which makes it a hidden file)
- 3 Add one pattern per line:
 - A folder name ending in `/` hides everything inside it
 - A `*` matches any characters (e.g., `*.pem` hides all `.pem` files)
 - Lines starting with `#` are comments
- 4 Save the file. Claude Code reads it automatically

Do this **before** your first Claude Code session.

If Claude Code has already read a file in the current conversation, adding it to `.claudeignore` later does not undo that.

The Three Execution Modes

The most important configuration decision you will make.

Plan Mode: Reviews approach with you *before* acting

Default Mode: Acts, but asks permission for “significant” steps

Auto-Accept: Executes everything without asking

↓ More autonomy, less oversight ↓

How it works:

- Claude Code analyzes your request
- Proposes a detailed plan (files to read, changes to make, approach)
- Waits for your explicit approval before acting
- You review, approve, modify, or reject

When to use:

Any task where the *approach* matters, not just the output.

Data transformation, variable construction, merges, anything analytical.

Why It Matters

The plan is your checkpoint. This is where you catch “I am going to do an inner join” *before* it happens, not after 300 observations are gone.

How it works:

- Claude Code executes steps autonomously
- Pauses for actions *it* considers significant
- Uses its own judgment about when to ask

When to use:

Routine tasks in established projects where you trust the general approach.

The Risk for Academics

Claude Code's judgment about what is “significant” may not match yours. It might pause before writing a file but **not** before silently recoding a variable.

How it works:

- Executes everything without asking
- Reads, writes, runs, and modifies freely
- Stops only on errors

When to use:

- File reformatting
- Boilerplate generation
- Documentation

When **not to use:**

Anything involving your data or analysis.

The speed gain is not worth the loss of checkpoints.

The Key Insight on Modes

Different Relationships with the Agent

Plan mode = you **direct** the work

Auto-accept = you **review** the work after the fact

For published research, directing is almost always safer than reviewing after the fact.

Why: Errors you *prevent* at the planning stage are cheaper than errors you *catch* in the output, or worse, errors you don't catch at all.

Live Demo: Plan Mode vs. Auto-Accept

Same task, two modes.

Watch for:

- What does a plan look like?
- How do you evaluate it?
- What does the experience feel like in each mode?

Choosing a Model

Claude Code lets you switch between three model tiers.

Each trades off **quality**, **speed**, **cost**, and **context window size**.

	Haiku 4.5	Sonnet 4.6	Opus 4.6
Context window	200k tokens	200k tokens	200k–1M tokens
Speed	Fastest	Fast	Slowest
Cost	Lowest	Moderate	Highest
Reasoning	Basic	Strong	Strongest

To switch models in Claude Code:

Type `/model` and select, or press Shift+Tab to toggle.

When to Use Each Model

Haiku

Fast, cheap, good enough for simple tasks.

- Reformatting files
- Renaming variables
- Generating boilerplate
- Quick questions about syntax

Sonnet

The everyday workhorse for most research tasks.

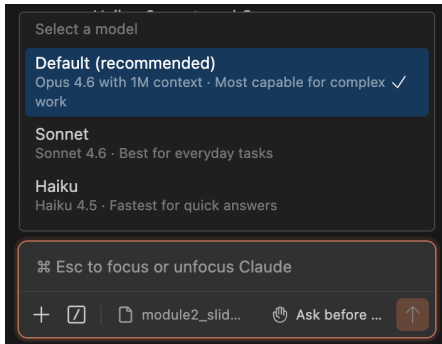
- Writing data cleaning scripts
- Building panels and merges
- Debugging code
- Drafting documentation

Opus

Maximum reasoning power; use when it matters.

- Complex multi-file refactors
- Subtle specification decisions
- Reviewing analytical code
- Large-context projects (up to 1M tokens)

Model Choice: The Research Perspective



The Key Tradeoff

Smarter models are slower and more expensive.
A wrong answer fast is worse than a right answer slow.

Rules of thumb:

- **Default to Sonnet:** best balance of quality and speed
- **Upgrade to Opus** for careful reasoning or many files
- **Drop to Haiku** for mechanical, easy-to-verify tasks

A Real Constraint

On the \$20/month plan, you can burn through your allocation in 30–45 minutes of active use if you are not strategic.

Principles:

- 1 Not everything needs Claude Code. Small edits are faster by hand
- 2 A well-written CLAUDE.md reduces back-and-forth (= fewer tokens)
- 3 If Claude Code is spinning on a wrong approach, **intervene early**
- 4 Batch related tasks into one well-scoped request
- 5 Plan mode adds tokens but prevents costly wrong approaches

Usage Limits and Model Cost

How limits work:

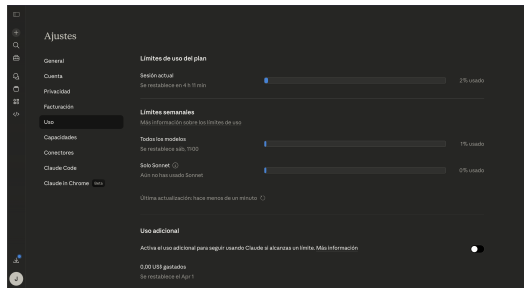
- Token allowance resets on a **5-hour rolling window**
- A separate **weekly cap** limits total usage
- When you hit either limit, Claude Code pauses until the window resets

Which models burn tokens fastest:

- **Opus** uses $\sim 5\times$ more tokens per task than Haiku
- **Sonnet** is the middle ground
- **Haiku** stretches your budget the furthest

Match the model to the task.

Don't use Opus to rename variables.



Where Claude Looks for Context

Claude Code reads instructions from **several levels** when it starts a session. Anything it finds is loaded into the context window before your first message.

- **User level** — `~/.claude/CLAUDE.md`

Applies to *every* project on your machine. Personal preferences, defaults, things you want Claude to know about you regardless of what you are working on.

- **Project level** — `CLAUDE.md` in the project root

Checked into git, shared with collaborators. Project conventions, data structure, constraints.

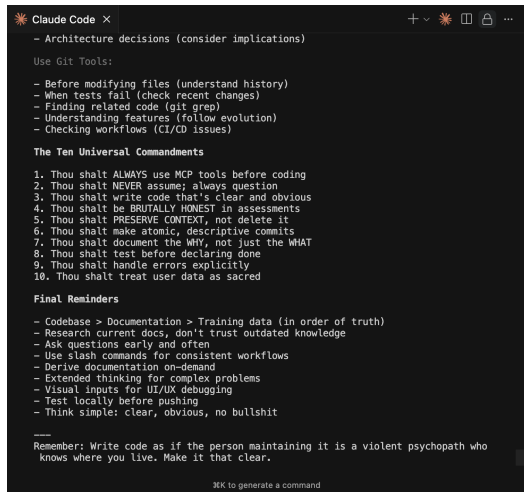
- **Subfolder level** — `CLAUDE.md` inside a subdirectory

Loaded when Claude works in that subtree. Useful for scoped rules (e.g. a `cleaning/` folder with its own conventions).

How Claude reads them

All applicable files are concatenated into context. More specific (deeper) files appear later, so they take precedence when guidance conflicts.

CLAUDE.md: Your Lab Notebook for the Agent



```
Claude Code X
- Architecture decisions (consider implications)

Use Git Tools:
- Before modifying files (understand history)
- When tests fail (check recent changes)
- Finding related code (git grep)
- Understanding features (follow evolution)
- Checking workflows (CI/CD issues)

The Ten Universal Commandments
1. Thou shalt ALWAYS use MCP tools before coding
2. Thou shalt NEVER assume; always question
3. Thou shalt write code that's clear and obvious
4. Thou shalt be BRUTALLY HONEST in assessments
5. Thou shalt PRESERVE CONTEXT, not delete it
6. Thou shalt make atomic, descriptive commits
7. Thou shalt document the WHY, not just the WHAT
8. Thou shalt test before declaring done
9. Thou shalt handle errors explicitly
10. Thou shalt treat user data as sacred

Final Reminders
- Codebase > Documentation > Training data (in order of truth)
- Research current docs, don't trust outdated knowledge
- Ask questions early and often
- Use slash commands for consistent workflows
- Derive documentation on-demand
- Extended thinking for complex problems
- Visual inputs for UI/UX debugging
- Test locally before pushing
- Think simple: clear, obvious, no bullshit

Remember: Write code as if the person maintaining it is a violent psychopath who knows where you live. Make it that clear.

⌘K to generate a command
```

Plain text files at the **user**, **project**, or **subfolder** level that Claude Code reads **automatically at the start of every conversation**.

They encode your conventions, definitions, and constraints so you do not have to repeat them each time.

The single most effective way to prevent errors before they happen.

Note: This does not guarantee Claude Code will always follow these instructions, but it significantly increases the chances.

Credit: u/rishabhsonker

What Belongs in CLAUDE.md

In the project-level CLAUDE.md, include:

- Project description & research question
- Data structure: files, key variables, units, expected ranges
- Coding conventions
- Constraints: “never drop observations without logging”
- “All merges must report row counts”
- What Claude Code should **not** do

Keep *user-level* CLAUDE.md for things that apply across projects; use a *subfolder* CLAUDE.md only when a directory needs its own rules.

How it interacts with the context window

Every applicable CLAUDE.md is loaded before your first message and stays “visible”—but it takes up space, so keep each one focused and concise.

CLAUDE.md Example

```
# CLAUDE.md - County Migration Project
```

```
## Research Question
```

```
Effect of historical migration on county-level outcomes.
```

```
## Data
```

- data/raw/census_panel.csv: county-year panel, 1920-2020
- Key vars: fips (5-digit), year, population, migration_share
- All monetary values in 2020 USD

```
## Constraints
```

- Never drop observations without logging count and reason
- All merges must report pre- and post-merge row counts
- Always use robust standard errors
- Do not choose model specifications
- Do not interpret coefficients

Exercise: Build the County Panel From Scratch

In Module 1 you ran a pre-built script. Now **you** build it.

Same data, fresh start. Work in pairs using plan mode.

- ❶ Write a CLAUDE.md for the project (use the template)
- ❷ Switch Claude Code to **plan mode**
- ❸ Ask Claude Code to build a county-year panel that merges election returns with IPUMS census demographics
- ❹ **Before approving the plan**, you will need to:
 - Read the codebooks to understand variable codings
(What does $\text{EDUC} \geq 10$ mean? What does $\text{RACE} == 1$ code?)
 - Decide how to define key variables
(Two-party vote share? College = bachelor's or higher?)
 - Specify the merge: which keys, which join type, expected row counts
- ❺ Review the plan, reject or revise as needed, then approve
- ❻ Run the script and verify the output

Evaluating the Plan

When Claude Code proposes its plan, ask yourself:

- Is it reading the **codebooks** or just guessing at variable meanings?
- How is it coding EDUC, RACE, HISPAN?
Do those match *your* definitions?
- What join type is it using? Inner, left, or outer?
What happens to counties that appear in one dataset but not the other?
- Is it filtering `mode == "TOTAL"` for election data?
- Does it handle `COUNTYFIP == 0` (unidentified counties)?

This is where your domain knowledge matters.

Claude Code can write the code, but you decide what the variables mean.

- What variable coding decisions did you make? Did Claude Code's default match yours?
- Did anyone reject or modify the plan? What did you change?
- How did your panel compare to the Module 1 version?

The Lesson

Writing a `CLAUDE.md` forces you to be explicit about conventions you have kept implicit.

Reviewing plans forces you to think about approach *before* you see output.

Domain knowledge is key for preventing errors.

- ① Claude Code works with **files**, not interactive sessions
- ② The **context window** limits what it can hold, so keep tasks focused
- ③ **Plan mode** is the default for research work
- ④ **CLAUDE.md** is your most powerful tool for preventing errors
- ⑤ **Cost management** is a core skill, not an afterthought

Next: Module 3, Automating Research Tasks